

Consumer Choice and the Demand Curve

Where a demand curve comes from

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Mankiw, *Principles of Microeconomics*

Chapter 21 — The Theory of Consumer Choice

Budget constraints, indifference curves, the consumer's optimum, and how demand curves follow from it.

One difference worth knowing. Mankiw works the chapter without calculus. These slides use it — marginal utilities as derivatives, the optimum by Lagrange — so what follows is the more general treatment, and the chapter is the gentler route into the same result.

Where we are going

1. Preferences and indifference curves
2. The budget constraint
3. The optimum — argued before it is solved
4. Solving it
5. A price change
6. Deriving the demand curve
7. Income and substitution effects
8. Why the substitution effect is never positive
9. What the model leaves out

Every number in the lecture comes from one utility function and four prices.

PART 1

Preferences and Indifference Curves

A bundle, and what we assume about it

Bundle. A list of quantities — so much of one good, so much of another. With two goods it is a pair of numbers, (x, y) .

Three assumptions, and they are all we will use:

1. **More is better.** A bundle with more of both is preferred.
2. **Completeness.** Any two bundles can be compared.
3. **Consistency.** If A is preferred to B and B to C , then A is preferred to C .

Nothing else is assumed about taste.

The utility function we will use

Utility. A number attached to a bundle, arranged so that a preferred bundle carries a higher number. Only the *ordering* means anything.

Throughout:

$$U(x, y) = x^{1/2} y^{1/2} = \sqrt{xy}$$

Feed it a bundle and it returns a number:

$$U(20, 30) = \sqrt{600} \approx 24.5, \quad U(10, 30) = \sqrt{300} \approx 17.3$$

The size of the number means nothing. That $24.5 > 17.3$ means everything.

Step 1 — write the indifference curve as an equation

Hold utility fixed at some level and ask which bundles deliver it.

$$\sqrt{xy} = \text{constant} \quad \implies \quad xy = k \quad \implies \quad y = \frac{k}{x}$$

Squaring both sides removes the root and costs nothing, because squaring does not disturb the ranking.

Through the bundle (20, 30): $k = 20 \times 30 = 600$, so

$$y = \frac{600}{x}$$

An indifference curve is the graph of an equation, not a shape drawn by feel.

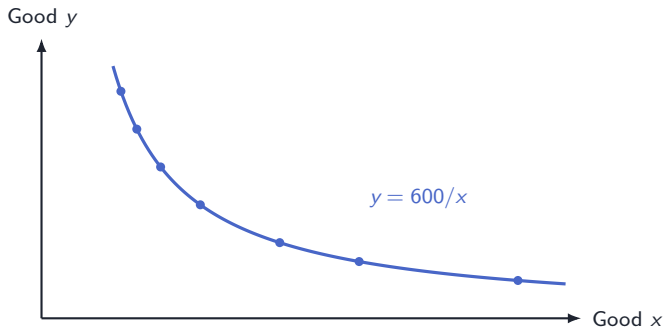
Step 2 — plug the numbers in

Every row divides 600 by x . Read across and multiply: the answer is 600 every time.

x	10	12	15	20	30	40	60
$y = 600/x$	60	50	40	30	20	15	10
xy	600	600	600	600	600	600	600

The consumer is **indifferent** between all seven. Each is $U = \sqrt{600}$.

Step 3 — draw the graph



Dots from the table, curve from the function — coinciding.

Two indifference curves cannot cross

Suppose two curves crossed at a bundle P .

- P is on the lower curve, so $U(P)$ is the lower utility.
- P is on the higher curve, so $U(P)$ is the higher utility.

One bundle, two different utilities. Impossible.

The argument uses nothing but consistency — assumption 3. It is the first time that assumption earns its keep, and not the last.

The slope: the marginal rate of substitution

Marginal rate of substitution. The amount of y the consumer will give up for one more unit of x , with utility held constant. It is the slope of the indifference curve, without its minus sign.

From $\rightarrow$ to	Δy	Δx	chord slope
$(10, 60) \rightarrow (20, 30)$	-30	+10	-3
$(20, 30) \rightarrow (40, 15)$	-15	+20	-0.75

The curve is flattening: a *diminishing* rate of substitution.

At a point the exact value is $MRS = y/x$, proved in Part 4.

Active Learning 1

Utility is $U(x, y) = \sqrt{xy}$. Consider the bundle $(12, 48)$.

1. Write the equation of the indifference curve through it.
2. Find the y that pairs with $x = 24$ on that same curve.
3. Compute the chord slope between the two bundles, and say what it tells you about the marginal rate of substitution.

Four minutes. Paper and pencil — the reasoning is the marks.

PART 2

The Budget Constraint

Step 1 — write the budget constraint

Preferences say what the consumer *wants*. The budget says what they *can have*.

Income \$120 a week. $p_x = \$3$. $p_y = \$2$.

Spending on x , plus spending on y , equals income:

$$\underbrace{3x}_{\text{spent on } x} + \underbrace{2y}_{\text{spent on } y} = \underbrace{120}_{\text{income}}$$

Every coefficient is a price. The right-hand side is income. Nothing here refers to taste.

Step 2 — plug the numbers in

Fix x , see what income is left, divide by $p_y = 2$.

x	0	8	16	20	24	32	40
spent on x : $3x$	0	24	48	60	72	96	120
left over: $120 - 3x$	120	96	72	60	48	24	0
$y = (120 - 3x)/2$	60	48	36	30	24	12	0

Every 4 units of x costs exactly 6 units of y — all the way along.

That constancy is why the line is straight.

Step 3 — rearrange to plot

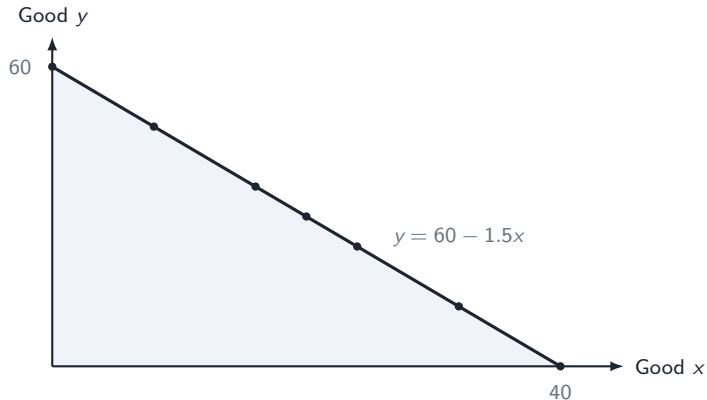
Make y the subject:

$$3x + 2y = 120 \implies 2y = 120 - 3x \implies y = 60 - 1.5x$$

- **60** is m/p_y — all the income spent on y .
- **1.5** is p_x/p_y — how much y each unit of x costs.

Nothing economic has happened. It is the same constraint, written the other way round.

Step 4 — draw the graph



Shaded: the **affordable set**. On the line: the whole income spent.

The slope is a relative price

The slope 1.5 is not a quantity of money. It is a **rate of exchange between goods**.

\$3 buys one unit of x .

\$3 buys one and a half units of y .

So the price of x , measured in y , is 1.5.

To get one more x , the market makes you give up 1.5 of y . That is an opportunity cost.

Active Learning 2

Income is \$120 and $p_y = \$2$, as before. But now $p_x = \$4$.

1. Write the budget equation.
2. Give both intercepts, and say which one moved and why.
3. Rearrange it to plot.
4. How much y must be given up for one more x ?

Three minutes. Paper and pencil.

PART 3

The Optimum

Which affordable bundle is best?

We now have two objects on the same pair of axes:

- a **budget line** — everything the consumer can afford;
- a **family of indifference curves** — how the consumer ranks bundles, higher curve meaning better.

The problem: find the affordable bundle on the highest curve it can reach.

The answer is on the line, never inside it. Money left unspent buys more of something, and more is better.

Plug every affordable bundle in

Substitute $y = 60 - 1.5x$ into xy and read down the column.

x	8	12	16	20	24	28	32
$y = 60 - 1.5x$	48	42	36	30	24	18	12
xy	384	504	576	600	576	504	384

It climbs to $x = 20$ and falls away — symmetrically.

The best affordable bundle is $(20, 30)$, and no calculus was used.

Why a non-tangency cannot be best — too much y

Take the affordable bundle (10, 45). Check: $3(10) + 2(45) = 30 + 90 = 120$. ✓

Consumer's rate: $MRS = y/x = 45/10 = \mathbf{4.5}$

Market's rate: $p_x/p_y = \mathbf{1.5}$

The consumer would hand over **4.5** units of y for one more x . The market charges **1.5**.

x is worth three times what it costs them. So they buy more x .

The same argument the other way — too much x

Now take $(30, 15)$. Check: $3(30) + 2(15) = 90 + 30 = 120$. ✓

Consumer's rate: $MRS = 15/30 = \mathbf{0.5}$

Market's rate: $p_x/p_y = \mathbf{1.5}$

The consumer will give up only **0.5** of y for another x . The market demands **1.5**.

Now x is dear relative to its worth. So they sell x and buy y .

Both ends push toward the middle.

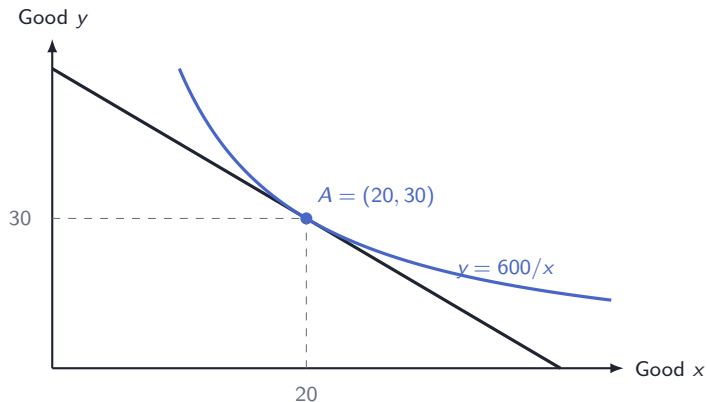
The tangency condition

$$\underbrace{MRS}_{\text{the consumer's rate}} = \underbrace{\frac{p_x}{p_y}}_{\text{the market's rate}} \quad \text{i.e.} \quad \frac{y}{x} = \frac{p_x}{p_y}$$

Here: $\frac{y}{x} = \frac{3}{2}$, and at (20, 30) that reads $\frac{30}{20} = 1.5$. ✓

The consumer's exchange rate and the market's agree. No trade is left that gains anything.

Draw the graph



They touch at one point and do not cross. That is tangency.

Active Learning 3

Same budget: $3x + 2y = 120$. Consider the bundle $(30, 15)$.

Predict first, before any arithmetic:

1. Is the *MRS* there above or below the price ratio?
2. Which way should the consumer move — more x , or less?

Then check yourself: compute xy at $(30, 15)$ and at $(20, 30)$, and say whether the check agrees with your prediction.

Four minutes. The reasoning is the marks.

PART 4

Solving It

Marginal utility, by differentiation

Marginal utility. The extra utility from one more unit of a good, with the other held fixed. That is a partial derivative.

With $U = x^{1/2}y^{1/2}$:

$$MU_x = \frac{\partial U}{\partial x} = \frac{1}{2}x^{-1/2}y^{1/2} = \frac{1}{2}\sqrt{\frac{y}{x}}, \quad MU_y = \frac{\partial U}{\partial y} = \frac{1}{2}x^{1/2}y^{-1/2} = \frac{1}{2}\sqrt{\frac{x}{y}}$$

Each falls as you get more of that good.

Diminishing marginal utility — a consequence of the function, not an extra assumption.

The rate of substitution, proved

Divide one marginal utility by the other:

$$MRS = \frac{MU_x}{MU_y} = \frac{\frac{1}{2}\sqrt{y/x}}{\frac{1}{2}\sqrt{x/y}} = \sqrt{\frac{y}{x}} \cdot \sqrt{\frac{y}{x}} = \frac{y}{x}$$

The $\frac{1}{2}$ s cancel. The exponents collapse. What is left depends only on the **ratio** y/x .

At (20, 30): $MRS = 30/20 = 1.5$ — which is what the chords in Part 1 were pointing at.

This is why Cobb–Douglas is tractable: the rate depends on the mix, never on the scale.

The Lagrangian

The problem:

$$\max_{x,y} x^{1/2}y^{1/2} \quad \text{subject to} \quad p_x x + p_y y = m$$

Build one function that carries both:

$$\mathcal{L}(x, y, \lambda) = x^{1/2}y^{1/2} + \lambda(m - p_x x - p_y y)$$

The bracket is **zero** at any bundle that spends the whole budget, so there $\mathcal{L} = U$.
Setting all three partial derivatives to zero solves the constrained problem.

The two first-order conditions

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{1}{2} \sqrt{\frac{y}{x}} - \lambda p_x = 0 \quad \implies \quad MU_x = \lambda p_x$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{1}{2} \sqrt{\frac{x}{y}} - \lambda p_y = 0 \quad \implies \quad MU_y = \lambda p_y$$

Each says: marginal utility equals λ times price.

Rearranged, $\frac{MU_x}{p_x} = \lambda = \frac{MU_y}{p_y}$ — the same utility per dollar in both goods.

Divide them

$$\frac{MU_x}{MU_y} = \frac{\lambda p_x}{\lambda p_y} \implies \frac{y}{x} = \frac{p_x}{p_y}$$

λ cancels — which is the whole point of dividing. It was never a quantity we cared about.

The tangency condition, argued in words in Part 3, falls straight out of the calculus.

The trading argument and the derivative give the same equation. That is the check: two routes, one answer.

Substitute into the budget — the closed forms

Tangency gives $y = \frac{p_x}{p_y}x$. Put that into $p_x x + p_y y = m$:

$$p_x x + p_y \left(\frac{p_x}{p_y} x \right) = m \quad \implies \quad p_x x + p_x x = m \quad \implies \quad 2p_x x = m$$

$$x^* = \frac{m}{2p_x}$$

$$y^* = \frac{m}{2p_y}$$

Two formulas, good for *any* income and *any* prices.

Plug the numbers in — bundle A

$$x^* = \frac{120}{2(3)} = \frac{120}{6} = \mathbf{20}, \quad y^* = \frac{120}{2(2)} = \frac{120}{4} = \mathbf{30}$$

Check	Arithmetic	
Budget constraint	$3(20) + 2(30) = 60 + 60 = 120$	✓
Tangency	$y/x = 30/20 = 1.5$ and $p_x/p_y = 3/2 = 1.5$	✓

Both conditions hold exactly — and this is the bundle the table in Part 3 found by exhaustion.

Half the budget on each good — whatever the prices

$$\text{Spending on } x \quad p_x x^* = 3 \times 20 = \$60$$

$$\text{Spending on } y \quad p_y y^* = 2 \times 30 = \$60$$

And that is not the numbers being kind:

$$p_x x^* = p_x \cdot \frac{m}{2p_x} = \frac{m}{2}$$

The price cancels. **Whatever** p_x is.

Equal exponents put half the budget on each good and keep it there.

Active Learning 4

A consumer has income \$200, faces $p_x = \$5$ and $p_y = \$4$, and has $U(x, y) = \sqrt{xy}$.

Find the best bundle **twice**:

1. from the closed forms;
2. from the tangency condition and the budget constraint, using no formula.

Then show the two routes agree, and check the budget balances.

Five minutes. If the routes disagree, you know to go looking.

PART 5

A Price Change

The price of x doubles

	before	after
p_x	\$3	\$6
p_y	\$2	\$2
income m	\$120	\$120

Predict before solving: x is dearer, so the consumer buys *less* of it.

The question is how much less — and the model answers exactly.

What moves, and what does not

New budget: $6x + 2y = 120$, so $y = 60 - 3x$.

	before	after	why
y-intercept m/p_y	60	60	neither m nor p_y moved
x-intercept m/p_x	40	20	\$120 now buys half as many x
slope p_x/p_y	1.5	3	each x now costs 3 of y

The line **pivots** about the point on the vertical axis. It does not shift.

Route one — the formula

$$x^* = \frac{m}{2p_x} = \frac{120}{12} = \mathbf{10}, \quad y^* = \frac{m}{2p_y} = \frac{120}{4} = \mathbf{30}$$

	before A	after B
x	20	10
y	30	30

The quantity of x halves. The quantity of y does not move at all.

Route two — tangency and budget, solved directly

No formula. Two equations, two unknowns.

Tangency: $\frac{y}{x} = \frac{p_x}{p_y} = \frac{6}{2} = 3, \text{ so } y = 3x.$

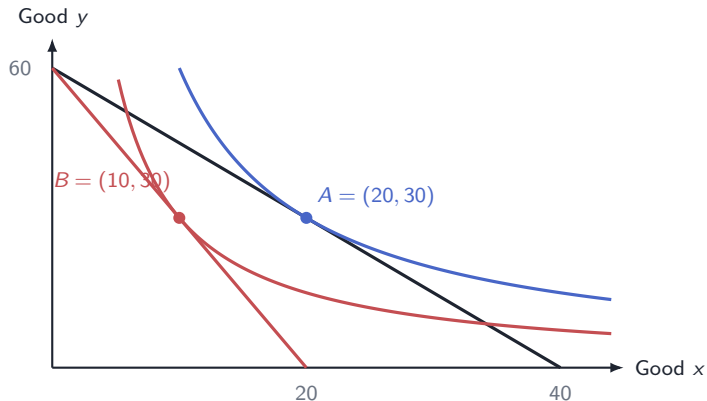
Budget: substitute it in —

$$6x + 2(3x) = 6x + 6x = 12x = 120 \quad \implies \quad x = 10, \quad y = 30$$

Budget check	$6(10) + 2(30) = 60 + 60 = 120 \checkmark$
Tangency check	$30/10 = 3 \text{ and } 6/2 = 3 \checkmark$

Both routes give (10, 30).

Draw the graph



The line pivots; the consumer lands on a **lower** curve ($k = 300$, was 600).

Spending on y did not move

	before	after	
spent on x	$3(20) = \$60$	$6(10) = \$60$	unchanged
spent on y	$2(30) = \$60$	$2(30) = \$60$	unchanged
quantity of y	30	30	unchanged

The formula says why: $y^* = \frac{m}{2p_y}$ — and p_x appears nowhere in it.

With this utility function, a change in the price of x leaves the market for y untouched.

That is a strong property. Part 9 returns to it.

Active Learning 5

Income \$120, $p_y = \$2$, $U = \sqrt{xy}$. Now p_x **falls** from \$3 to \$2.

Predict first, in writing:

1. Which way does the quantity of x move?
2. Which way does the quantity of y move?

Then solve: find the new bundle and check it against the budget constraint. Explain the answer to (2).

Four minutes.

PART 6

Deriving the Demand Curve

Do it for every price

We solved the problem at $p_x = \$3$ and at $p_x = \$6$. There is no reason to stop.

$$x^* = \frac{m}{2p_x} \quad \text{with } m = 120 : \quad \boxed{x^* = \frac{60}{p_x}}$$

One equation. Every price at once. That is a demand function.

Nothing new is assumed here. It is the Part 4 result with the income written in.

Step 2 — plug the numbers in

p_x	\$1	\$2	\$3	\$4	\$5	\$6
$x^* = 60/p_x$	60	30	20	15	12	10
spent: $p_x x^*$	\$60	\$60	\$60	\$60	\$60	\$60

Every entry in the middle row is 60 divided by the price above it — check any of them.

And the bottom row never changes. Hold on to that.

Step 3 — rearrange to plot

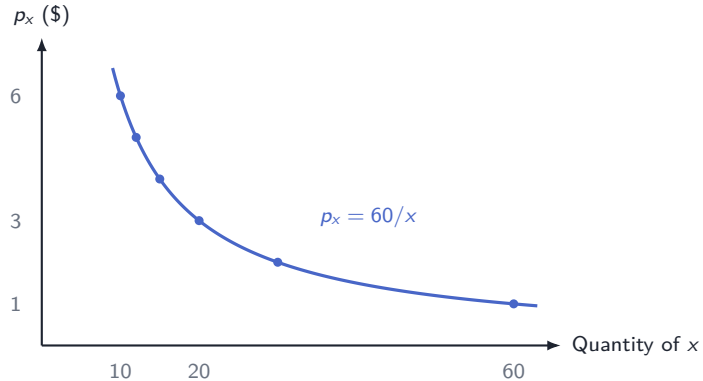
Economists put price on the *vertical* axis, so make p_x the subject:

$$x^* = \frac{60}{p_x} \quad \Longleftrightarrow \quad p_x = \frac{60}{x}$$

The same relationship, written the other way round.

Both forms say $p_x \cdot x = 60$. It is a **rectangular hyperbola** — the same curve family as an indifference curve, on different axes and for a completely different reason.

Step 4 — draw the graph



Dots from the table, curve from the function — coinciding.

What has just happened

Before: a demand curve was written down, and assumed to slope downward because that seemed sensible.

Here: nothing about demand was assumed at all.

- preferences $U = \sqrt{xy}$
- a budget $p_x x + p_y y = m$
- do the best you can

Out came: $x^* = 60/p_x$.

The demand curve is a *result*, and we know its exact equation.

Expenditure is \$60 at every price

Price $\times$ quantity, from the table: $1(60) = 2(30) = \dots = 6(10) = \60 .

Constant expenditure is exactly what **unit elastic** means. Check it with the midpoint method, p_x from \$3 to \$6:

$$\% \Delta Q = \frac{10 - 20}{15} = -\frac{2}{3}, \quad \% \Delta P = \frac{6 - 3}{4.5} = +\frac{2}{3}$$

$$\text{elasticity} = \frac{-2/3}{+2/3} = -1$$

Exactly minus one — and it comes out at minus one for *every* pair of prices on this curve.

Active Learning 6

Using $x^* = 60/p_x$:

1. Compute the quantity at $p_x = \$10$ and at $p_x = \$12$, and show what is spent on x in each case.
2. Apply the midpoint method to that pair.
3. Apply it again to the pair $p_x = \$1$ and $p_x = \$2$.
4. Say why the two elasticities come out the same.

Four minutes. Part (4) is where the marks are.

PART 7

Income and Substitution Effects

The fall from 20 to 10 is two things at once

When p_x doubled, x fell from 20 to 10. **Two separate causes.**

1. x became dearer *relative to* y , so the consumer swings toward y .
A change in the rate of exchange.
2. The consumer became *poorer* — \$120 no longer buys what it did.
A change in what is affordable at all.

To separate them, remove the second one — then whatever is left is the first.

Compensate: keep the old bundle just affordable

Slutsky compensation. Give the consumer exactly enough income that the *original bundle* is affordable at the *new* prices.

Cost bundle $A = (20, 30)$ at the new prices:

$$m' = p_x^{\text{new}} x_A + p_y y_A = 6(20) + 2(30) = 120 + 60 = \text{\$180}$$

At \$180 the consumer *could* buy exactly what they bought before. So nothing has been taken away.

Only the relative price has changed.

The compensated bundle C

Solve again — new prices, compensated income $m' = \$180$:

$$x_C = \frac{m'}{2p_x} = \frac{180}{12} = \mathbf{15}, \quad y_C = \frac{m'}{2p_y} = \frac{180}{4} = \mathbf{45}$$

Budget check	$6(15) + 2(45) = 90 + 90 = 180 \checkmark$
Tangency check	$y/x = 45/15 = 3 \quad \text{and} \quad p_x/p_y = 6/2 = 3 \checkmark$

The compensated budget line is $y = 90 - 3x$: same slope as the new line, further out.

The split

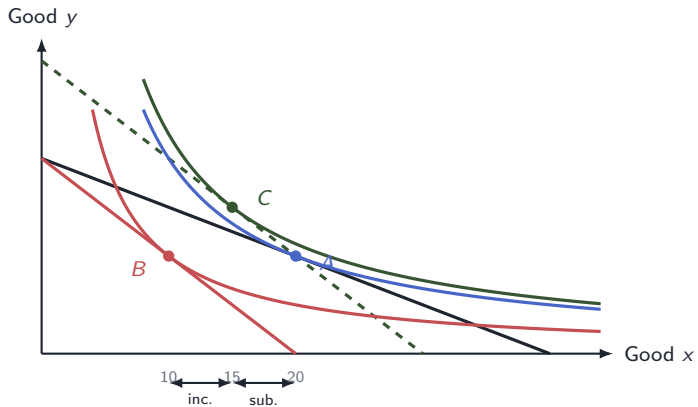
$$\begin{array}{ccc} \underbrace{20 \rightarrow 15} & \underbrace{15 \rightarrow 10} & \underbrace{20 \rightarrow 10} \\ \text{substitution: } -5 & \text{income: } -5 & \text{total: } -10 \end{array}$$

- $A \rightarrow C$: relative price changes, purchasing power held fixed.
- $C \rightarrow B$: the compensation is taken away again, prices unchanged.

−5 and −5 sum to −10, the whole fall. They must.

That the two are *equal* here is a property of this utility function, not a law.

Draw the graph



The dashed line passes exactly through A — that is what compensation means.

The compensation was generous

Which indifference curve does each bundle sit on?

$$k_A = 20(30) = 600, \quad k_C = 15(45) = \mathbf{675}, \quad k_B = 10(30) = 300$$

$675 > 600$: *C is on a higher curve than A.*

Keeping the old *bundle* affordable is more generous than keeping *utility* unchanged — because the consumer does not buy the old bundle. They buy something better.

Active Learning 7

Income \$240, $p_y = \$4$, $U = \sqrt{xy}$. The price of x rises from \$3 to \$4.

1. The original bundle and the final bundle.
2. The Slutsky-compensated income.
3. The compensated bundle.
4. The substitution effect and the income effect.
5. Check that they sum to the total change.

Five minutes. Show every substitution.

PART 8

Why the Substitution Effect Is Never Positive

The claim

Raise the price of a good and hold purchasing power fixed by compensating.
The consumer never buys *more* of it.

The proof uses:

- no calculus;
- no assumption about the shape of preferences;
- nothing about this particular utility function.

Only that the consumer **chooses consistently**.

The two facts we already have

Fact 1. Under the compensated budget, the old bundle A is exactly affordable.

That is what the compensation was constructed to do: $m' = p'_x x_A + p_y y_A$ is precisely the cost of A at the new prices.

Fact 2. Facing that budget, the consumer chose C — not A .

A was available and was passed over.

So C is at least as good as A .

Suppose the opposite

Suppose the result is false. Suppose the compensated bundle C contains *more* x than A did:

$$x_C > x_A$$

We follow that supposition until it breaks.

Nothing else changes. C is still on the compensated budget line, and A is still just affordable on it.

Cost the two bundles at the *old* prices

Both cost m' at the new prices, so extra spending on x must be matched by less on y . Subtracting the two budget identities gives:

$$\underbrace{(p_x x_C + p_y y_C)}_{\text{cost of C, old prices}} - \underbrace{(p_x x_A + p_y y_A)}_{\text{cost of A, old prices}} = (p_x - p'_x)(x_C - x_A)$$

Here $p_x - p'_x = 3 - 6 = -3$.

If $x_C > x_A$, the product is **negative**: C would have been strictly cheaper than A at the old prices.

The contradiction

At the old prices: C was affordable, with money to spare — and the consumer chose A .

$\implies A$ is preferred to C

Under the compensated budget: A was affordable — and the consumer chose C .

$\implies C$ is preferred to A

Both cannot hold. The supposition fails, so $x_C \leq x_A$ always.

The compensated demand curve slopes down — for *any* consistent consumer.

Ordinary demand can still slope up

The theorem is about **compensated** demand. Ordinary demand contains the income effect too:

$$\underbrace{\text{total}}_{\text{ordinary demand}} = \underbrace{\text{substitution}}_{\text{never positive}} + \underbrace{\text{income}}_{\text{either sign}}$$

If the income effect is positive and *large enough* to outweigh the substitution effect, ordinary demand slopes **upward**.

Such a good is called a Giffen good.

They are rare, and hard to find convincingly in real data.

Active Learning 8

At $p_x = \$3$, $p_y = \$2$, income \$120, a consumer chose (20, 30).

The price of x then rose to \$6 and income was compensated to \$180. The consumer was observed choosing **(25, 15)**.

Is that observation consistent with a consumer who chooses consistently?

Show the arithmetic that settles it.

Five minutes. No formula will do this for you.

PART 9

What the Model Leaves Out

Cobb–Douglas is a special case

Our utility function has a name — **Cobb–Douglas** — and it is convenient rather than general.

- Constant expenditure shares: always half on each good.
- Therefore unit-elastic demand: always exactly -1 .
- Therefore **no cross-price effect**: p_x can do what it likes and the quantity of y never moves.

So the two goods cannot be substitutes or complements in any interesting way.

Preferences are given, stable and consistent

The model takes preferences as **data**. It says nothing about:

- where they come from;
- whether advertising moves them;
- whether they change with age, mood, or company;
- whether the consumer knows what is available, or what things cost.

That is not a small thing to assume. The whole apparatus rests on it.

Two goods

The page has two dimensions, so the picture does.

But **nothing in the logic** depends on it. The tangency condition and the closed forms extend to any number of goods:

$$\frac{MU_1}{p_1} = \frac{MU_2}{p_2} = \dots = \frac{MU_n}{p_n}$$

What is lost with more goods is the graph, not the argument.

Slutsky, not Hicks

We compensated to keep the *old bundle affordable*. That is **Slutsky**.

The alternative keeps *utility unchanged*. That is **Hicks**, and it hands over less money — about \$170 here rather than \$180, because we saw that \$180 left the consumer slightly better off.

Both give a substitution effect that is **never positive**. The magnitudes differ slightly.

A model whose limits you cannot state is a model you do not understand.

Four limitations were just listed:

- (a) Cobb–Douglas is a special case
- (b) preferences given and stable
- (c) two goods
- (d) Slutsky rather than Hicks

You want to use this model to study a consumer choosing between coffee and tea. Which limitation is the most damaging, and why?

Three minutes. Two or three sentences — the reasoning is the marks.

A demand curve is not an assumption.

It is what a consumer with preferences and a budget *does* when they do the best they can — one answer for every price, and the collection of those answers is the curve.

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